

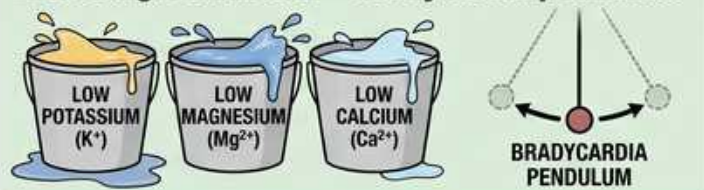
Arrhythmias — Long QT & Torsades de Pointes (Torsades Twister)

SCENE 3 — Long QT & Torsades: "The Torsades Twister" · arr_longqt_torsades.png

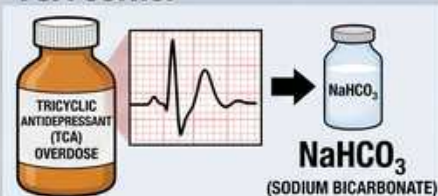
Pharmacy wall of QT-prolongers



Low K/Mg/Ca buckets + bradycardia pendulum



TCA corner



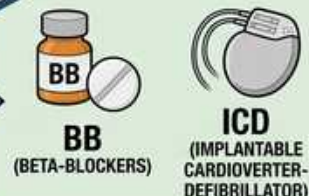
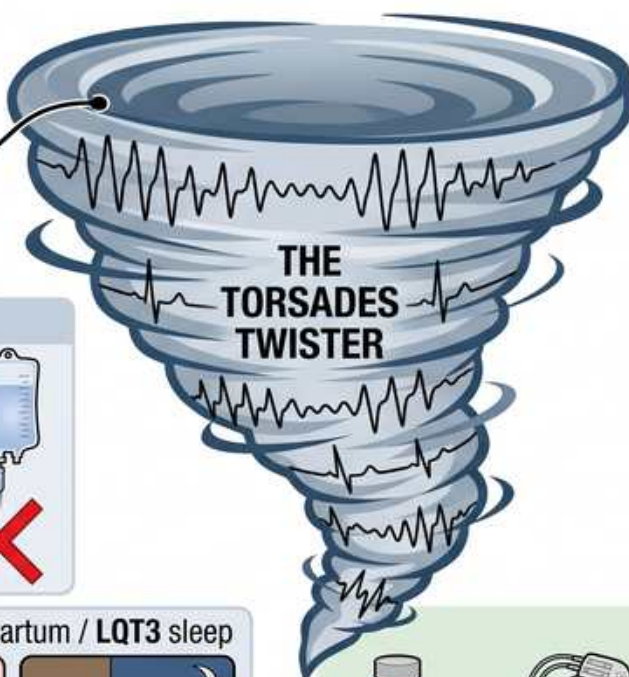
Congenital: LQT1 swim / LQT2 auditory-postpartum / LQT3 sleep



Stretched QT clock



Tx (Treatment)



1. QT-prolonging drug wall

WHAT YOU SEE

Pharmacy wall: class IA, sotalol/dofetilide, azithromycin, haloperidol, amitriptyline, ondansetron, chloroquine, methadone.

WHAT IT ENCODES

Acquired long QT is most often drug-induced. High-yield classes: class IA (quinidine/procainamide/disopyramide) and class III (sotalol, dofetilide, ibutilide), macrolides (azithromycin), fluoroquinolones, antipsychotics (haloperidol), TCAs (amitriptyline), antiemetics (ondansetron), antimalarials (chloroquine), methadone. Risk compounds with multiple QT drugs and electrolyte depletion.

BOARD TRAP

WRONG: 'amiodarone causes the most torsades.' Amiodarone markedly prolongs QT but rarely causes TdP; sotalol and the listed agents are higher-risk for TdP.

2. Digoxin does NOT prolong QT

WHAT YOU SEE

Digoxin bottle, green: 'does NOT prolong QT'.

WHAT IT ENCODES

Digoxin is the classic distractor — it SHORTENS the QT (with the 'Salvador Dalí mustache' ST scooping) and does not cause torsades. Its toxicity is bidirectional/atrial tachycardias and AV block, not TdP.

BOARD TRAP

WRONG: 'add digoxin to the avoid-in-long-QT list.' Digoxin shortens QT — it is the safe option among QT distractors.

3. Low K/Mg/Ca buckets

WHAT YOU SEE

Three depleted buckets: low potassium, magnesium, calcium.

WHAT IT ENCODES

Hypokalemia, hypomagnesemia, and hypocalcemia all prolong repolarization and predispose to torsades. Correcting them — especially magnesium — is both prevention and treatment.

BOARD TRAP

WRONG: 'electrolytes don't matter if the QT drug is stopped.' Repleting K and Mg is essential even after withdrawing the offending drug.

4. Bradycardia / pause pendulum

WHAT YOU SEE

Slow pendulum = pause-dependent torsades.

WHAT IT ENCODES

Acquired (pause-dependent) torsades is triggered by bradycardia and post-pause QT lengthening. Treat refractory cases by raising the rate — overdrive pacing or isoproterenol — to shorten the QT.

BOARD TRAP

WRONG: 'slow the heart rate to suppress torsades.' Bradycardia worsens pause-dependent TdP; you increase the rate.

5. Stretched QT clock

WHAT YOU SEE

A clock stretched long.

WHAT IT ENCODES

QTc >500 ms (or a >60 ms increase from baseline) markedly raises torsades risk. Measure QT in lead II/V5, correct for rate (Bazett), and stop the offending drug when prolonged.

BOARD TRAP

WRONG: 'absolute QT matters, not rate-corrected.' Use the rate-corrected QTc; a long absolute QT at slow rates may be normal.

6. Torsades twister (polymorphic VT)

WHAT YOU SEE

A tornado of twisting polymorphic VT.

WHAT IT ENCODES

Torsades de pointes is a polymorphic VT with a twisting axis around the baseline, arising on a prolonged QT. It may self-terminate (syncope) or degenerate into VF.

BOARD TRAP

WRONG: 'treat torsades like monomorphic VT with amiodarone.' Amiodarone prolongs QT and can worsen TdP — give magnesium instead.

7. Treatment: stop drug + IV magnesium

WHAT YOU SEE

Tx box: STOP DRUG, IV magnesium.

WHAT IT ENCODES

First steps in torsades: STOP the offending drug, give IV magnesium sulfate (even if Mg is normal), correct K to high-normal, and treat bradycardia/pauses with pacing or isoproterenol. Unstable/sustained TdP → defibrillation.

BOARD TRAP

WRONG: 'load amiodarone for recurrent torsades.' Magnesium ± overdrive pacing is the answer; QT-prolonging antiarrhythmics are avoided.

8. AVOID amiodarone

WHAT YOU SEE

Amiodarone crossed out.

WHAT IT ENCODES

QT-prolonging antiarrhythmics (amiodarone, sotalol, procainamide) are avoided in torsades because they lengthen the QT further and can perpetuate the arrhythmia.

BOARD TRAP

WRONG: 'amiodarone is the antiarrhythmic of choice for any VT.' Not for torsades/long-QT — it makes the substrate worse.

9. TCA overdose → NaHCO₃

WHAT YOU SEE

TCA corner: wide QRS → sodium bicarbonate.

WHAT IT ENCODES

Tricyclic antidepressant overdose blocks fast sodium channels → wide QRS, hypotension, and arrhythmia; IV sodium bicarbonate is the treatment (narrows QRS, overcomes Na-channel block). TCAs also prolong QT.

BOARD TRAP

WRONG: 'give a class IA antiarrhythmic for TCA-related wide-complex rhythm.' Class IA worsens Na-channel block — use sodium bicarbonate.

10. Congenital LQT1/2/3 triggers

WHAT YOU SEE

Swim (LQT1), auditory/postpartum (LQT2), sleep (LQT3) figures.

WHAT IT ENCODES

Congenital LQTS subtypes have classic triggers: LQT1 (KCNQ1) — exercise/swimming; LQT2 (KCNH2) — emotion/auditory startle/postpartum; LQT3 (SCN5A) — rest/sleep. β -blockers are mainstay (best in LQT1/2).

BOARD TRAP

WRONG: 'sudden death during sleep points to LQT1.' Rest/sleep events suggest LQT3; exertional/swimming events suggest LQT1.

11. β -blocker + ICD

WHAT YOU SEE

BB bottle and an ICD device.

WHAT IT ENCODES

Long-QT management: β -blockers for most; add an ICD for survived cardiac arrest, recurrent syncope on β -blockers, or very high risk (e.g., QTc >500 ms with features). Avoid all QT-prolonging drugs.

BOARD TRAP

WRONG: 'ICD first-line for all congenital long QT.' β -blockers are first-line; ICD is reserved for high-risk/secondary prevention.
